Fiscal Year:	FY 2016	**Task Last Updated:**	FY 10/27/2015
PI Name:	Minna, John D. M.D.		
Project Title:	NSCOR: Risks Estimates and Mechanisms of Lung Cancer Pathogenesis after Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	john.minna@utsouthwestern.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	214 648-4900
Organization Name:	The University of Texas Southwestern Medical Center		
PI Address 1:	Hamon Center for Therapeutic Oncology Research		
PI Address 2:	6000 Harry Hines Boulevard, NB8.206		
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75390-8593	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiation NSCOR/Virtual NSCOR NNJ10ZSA002N
Start Date:	01/01/2011	**End Date:**	12/31/2016
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2016 per S. Monk/SR-LaRC (Ed., 10/20/15)		
Key Personnel Changes/Previous PI:	October 2015 report: David Chen, previously Lead for project 3, retired from The University of Texas Southwestern Medical Center. Aroumougame Asaithamby, who was already Co-Lead for project 3 assumed the Lead role until the completion of the project. Allan Balmain was added as the Lead for the new project 3. Rolf Brekken was added as the Lead for the new project 4.		
COI Name (Institution):	Aroumougame, Asaithamby Ph.D. (The University of Texas Southwestern Medical Center) Gazdar, Adi F. M.D. (The University of Texas Southwestern Medical Center) Girard, Luc Ph.D. (The University of Texas Southwestern Medical Center) Shay, Jerry William Ph.D. (The University of Texas Southwestern Medical Center) Story, Michael Ph.D. (The University of Texas Southwestern Medical Center) Xie, Xian-Jin (The University of Texas Southwestern Medical Center) Xie, Yang Ph.D. (The University of Texas Southwestern Medical Center) Brekken, Rolf Ph.D. (The University of Texas Southwestern Medical Center) Balmain, Allan Ph.D. (University of California San Francisco)		
Grant/Contract No.:	NNX11AC54G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	This NASA Specialized Center of Research (NSCOR) focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced initiation and promotion events. They also measure effects on development of cancer stem-like cells (cancer initiating cells). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, University of Texas Southwestern Medical Center (UTSW) has assembled a team of scientists (Drs. John Minna, Jerry Shay, Michael Story, Yang Xie, Luc Girard, Adi Gazdar, D. Chen, and A. Asaithamby) who are leaders in the study and translation application of the molecular pathogenesis of lung cancer and in radiobiology. We have a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized human bronchial epithelial cell (HBEC) system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified and that specific mRNA expression profiles for HZE irradiation exist. In addition, we have found that transgenic mouse models of lung cancer have increased lung cancer risk after high LET (linear energy transfer) exposure allowing us to study the early onset and progression of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. HZE Particle Exposure and the Risk for Human Lung Carcinogenesis; 2. Mouse Models of Lung Cancer after HZE Particle Irradiation; 3. Organotypic (3D) Human Models of Genomic Instability for HZE Particle Induced Lung Carcinogenesis; 4. Integrating Biomarkers of Lung Cancer Pathogenesis for Risk Assessment after HZE Particle Irradiation. These projects are supported by 2 Cores: Administrative Core and the Database, Biostatistics and Bioinformatics Core. Our NSCOR has substantial UTSW institutional commitments, a recently renewed National Cancer Institute (NCI) Special Program of Research Excellence (SPORE) in Lung Cancer, a prior Center for the NCI Early Detection Research Network (EDRN) program, and key investigators that are part of other funded NSCORs (Shay and Story). These provide great synergism and additional resources available for the successful completion of this renewal proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused mostly by smoking exposure, a significant number of lung cancers arise in life time of never smokers and thus is a sizable health problem in the USA. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens, especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. In addition, in the past decade in the USA, Europe, and Asia there has been a resurgent interest in radiographic screening for lung cancer, particularly using serial computed tomography (CT) scans. In fact, the USA NCI has completed a large (50,000 person) National Lung Cancer Screening Trial (NLST) that randomized persons to have CT scans or not for the early detection of lung cancer. This trial has found a significant survival benefit for persons receiving CT scan screening. This raises a variety of questions concerning the use of this approach because of its cost and substantial population based radiation exposure. Greater than 25% of persons screened (current or former smokers) will have some lesions requiring further CT scan follow-up; when biopsies are performed only 1% of the overall group will have histologic evidence of lung cancer. In addition, many of the persons will require several CT scans in follow-up to determine if the lesion is growing to warrant a biopsy. Thus, there is considerable interest in knowing the contribution of CT scan doses of gamma radiation to the pathogenesis of lung cancer as well as other environmental exposures. This is particularly true since this radiation is being administered to individuals (current and former smokers) who are at highest risk of having preneoplastic lesions leading to lung cancer. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation still need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 50 different individuals) and innovative genetically engineered mouse models (GEMMs) of lung cancer comparing cells and mice before and after space radiation (performed at Brookhaven National Laboratory). Our studies are providing quantitative data following high-LET and low dose gamma irradiation on these HBECs and the GEMMs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high-LET radiation in space in order to reduce the uncertainties in prediction of risk of astronauts developing lung cancer as part of long-term missions in space. Thus, the results will also have major impact and benefit to life on Earth that includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through CT scan screening studies, and environmental exposures, accidentally or through terrorism. We are determining quantitative changes in biologic phenotypes related to progression toward malignancy of lung epithelial cells in response to radiation, which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional (3D) cultures, as well as growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA, and protein expression, and signaling pathway changes.

Task Progress:

We have made significant progress on the effects for different types and doses of space radiation on human cells and mouse models. In addition, we have major functional/mechanistic discoveries which include our data demonstrating that:
1. The relative biologic effective (RBE) doses of HZE particle radiation for human lung epithelial cells (HLECs) for a variety of ions and energies include Fe, Si, O, H;

2. HLECs can be partially transformed to malignancy by space radiation and we identified the dose range (0.25-0.4 Gy, for Si and Fe at 1000 MeV/n) where transformation is maximal falling off at lower and higher doses;

3. HLECs are also very resistant to full malignant transformation to lung cancer;

4. There is inter-individual variation in HLEC malignant transformation induced by HZE particle radiation which is only found at low dose rates (0.2 cGy/min) that are most relevant for space flight but not by high doses;

5. There are common and unique mRNA expression signatures in HLECs and mouse lung following HZE particle irradiation that predict the radiation types these cells were exposed to (molecular memory of the type of their prior HZE particle exposure);

6. In HLECs with pre-existing oncogenic changes (premalignancy, such as with p53 and KRAS abnormalities), there is earlier and quantitatively greater malignant transformation (soft agar colony growth) after HZE particle radiation and, also, a shift to a mesenchymal phenotype with accompanying increase in radioresistance, and shift in the transformed cells to a state of oxidative stress independent of mitochondrial proton leakage;

7. Mice susceptible to lung cancer will develop more invasive lung cancers after protracted/fractionated space radiation, including simulation of solar particle events, compared to terrestrial radiation;

8. That the development of space radiation induced cancer (SRIC) in mouse models is associated with micro-environmental inflammatory changes is highly suggestive of an effect of space radiation on the host microenvironment such as the induction of immune tolerance;

9. Changes in mRNA expression (mRNA signatures) are found in the mouse lung, long after space radiation but before cancers develop. These same signatures when found in human tumors, provide prognostic information for resected human lung and breast cancers. Thus, the same gene expression changes indicating inflammatory signaling after HZE particle radiation occur in human tumors with a worse prognosis after surgical resection;

10. Development of methods to grow HLECs in 3D culture (more physiologically relevant) which we then were able to study and find such 3D cultures exhibited shut down of DNA repair gene expression with down regulation of the DNA repair gene RAD51 playing a major role in genomic stability after space radiation of HLECs leading to chromosomal instability. We found that repair of the complex double strand breaks (DSBs) induced by high-LET iron particles, but not the isolated DSBs induced by gamma-rays, are substantially compromised by 3D organotypic culture, which appears related to decreases in RAD51, that is critical for homologous recombination repair of DNA double strand breaks. Thus, HZE particle radiation may have a greater effect when delivered in tissue-like conditions. This system also allows integration and study of microenvironmental changes into the 3D in vitro system.

11. We have a variety of findings related to identifying lung cancer stem cells that will have great importance for understanding HZE particle radiation induced lung cancer;

12. Using systems biology and bioinformatics we have integrated molecular data following HZE particle and gamma-radiation to develop risk assessment models for increases in fatal lung cancers from exposure to space radiation. These led to developing new tools for systems biology analyses, and identified a small number of master regulating hub genes which predict expression changes in many other genes after various types of radiation, which we are currently evaluating.

New Directions: Being our last year of this funded NSCOR cycle (ends December 2015) [Ed. note: 11/2/15--NSCOR has been extended to December 2016]. we have submitted a renewal proposal to NASA and have made some personnel changes (with the retirement from UTSW of Dr. David Chen).

Collaboration with University of California at San Francisco (UCSF) NSCOR Investigators: We believe it is now well established that the types of DNA damage are more complex upon exposure to HZE ions compared to gamma irradiation and have published several papers on this topic. However, we decided that a more global omics approach would be necessary going forward. As such we established a collaboration with UCSF and Dr. Allan Balmain who has prior NSCOR support and has a very strong mouse oriented approach using radio-sensitive and resistant strains. Dr. Balmain has developed a novel delta-TP53 mouse model that has a very low background rate of developing malignancy, but then reproducibly develops tumors at several different sites after radiation including HZE particle radiation. In addition, he has uniquely studied inter-crosses between Mus spretus and more typical inbred strains of mice (Mus musculus) that will permit us to identify radiation quality effects on the genomic landscape after high and low LET irradiation. Our combined (UTSW and UCSF) team and current NSCOR progress will provide the first comprehensive dataset on the role of space radiation induction of cancers of various tissues including lung in intact mice.

Study of development of immune tolerance after space radiation: An important part of the space radiation induced changes in the mouse models we have observed is the induction of inflammatory gene expression signatures long after the radiation, but before tumor induction indicating the need to study the host/tumor microenvironment and tumor immune responses which we are proposing going forward with our studies. In addition, over the past 2-5 years, major new discoveries of the role of immune tolerance and breaking such tolerance in the pathogenesis and treatment of human lung cancer have been made. Because of all of these findings, we are initiating studies on assessing the effects of space radiation on developing immune tolerance to cancer. We have enlisted Dr. Rolf Brekken a distinguished UTSW professor of surgical oncology and pharmacology to address this key issue of identifying acquired changes in immune responses following high and low LET radiation compared to no radiation. This project will work closely with all other projects in our NSCOR to define the functional importance of radiation induced omic changes of immune tolerance, which should provide a mechanistic understanding of cancer risk and potential countermeasure targets. This new project will include identifying common immunologic themes that occur in different mouse models of cancer; the role of tumor cell autonomous molecular changes for developing immune tolerance; and the specific role of exposed phosphatidylserine as a biomarker and potential target for immune checkpoint blockade countermeasures.

Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Ding LH, Park S, Xie Y, Girard L, Minna JD, Story MD. "Elucidation of changes in molecular signalling leading to increased cellular transformation in oncogenically progressed human bronchial epithelial cells exposed to radiations of increasing LET." Mutagenesis. 2015 Sep;30(5):685-94. Epub 2015 May 22. http://dx.doi.org/10.1093/mutage/gev028 ; PubMed PMID: 26001755 , Sep-2015
Articles in Peer-reviewed Journals	Zhong R, Dong X, Levine B, Xie Y, Xiao G. "iScreen: Image-based high-content RNAi screening analysis tools." J Biomol Screen. 2015 Sep;20(8):998-1002. Epub 2014 Dec 29. http://dx.doi.org/10.1177/1087057114564348 ; PubMed PMID: 25548139 , Sep-2015
Articles in Peer-reviewed Journals	Camacho CV, Todorova PK, Hardebeck MC, Tomimatsu N, Gil del Alcazar CR, Ilcheva M, Mukherjee B, McEllin B, Vemireddy V, Hatanpaa K, Story MD, Habib AA, Murty VV, Bachoo R, Burma S. "DNA double-strand breaks cooperate with loss of Ink4 and Arf tumor suppressors to generate glioblastomas with frequent Met amplification." Oncogene. 2015 Feb 19;34(8):1064-72. Epub 2014 Mar 17. http://dx.doi.org/10.1038/onc.2014.29 ; PubMed PMID: 24632607; PubMed Central PMCID: PMC4167163 , Feb-2015
Articles in Peer-reviewed Journals	Niwa O, Barcellos-Hoff MH, Globus RK, Harrison JD, Hendry JH, Jacob P, Martin MT, Seed TM, Shay JW, Story MD, Suzuki K, Yamashita S. "Stem Cell Biology with Respect to Carcinogenesis Aspects of Radiological Protection." Annals of ICRP (International Commission on Rdiological Protection). 2015: (In Press as of October 2015). Draft report dated July 2014. , Oct-2015
Articles in Peer-reviewed Journals	Adams CJ, Yu JS, Mao JH, Jen KY, Costes SV, Wade M, Shoemake J, Aina OH, Del Rosario R, Menchavez PT, Cardiff RD, Wahl GM, Balmain A. "The Trp53 delta proline (Trp53DeltaP) mouse exhibits increased genome instability and susceptibility to radiation-induced, but not spontaneous, tumor development." Mol Carcinog. 2015 Aug 27. [Epub ahead of print] http://dx.doi.org/10.1002/mc.22377 ; PubMed PMID: 26310697 , Aug-2015
Articles in Peer-reviewed Journals	Kim SB, Zhang L, Shay JW. "Oxygen and silicon ion particles induce neoplastic transformation in human colonic epithelial cells." Gravit Space Res. 2014 Aug;2(1):32-41. http://gravitationalandspacebiology.org/index.php/journal/article/view/640 , Aug-2014
Articles in Peer-reviewed Journals	Shao C, Sullivan JP, Girard L, Augustyn A, Yenerall P, Rodriguez-Canales J, Liu H, Behrens C, Shay JW, Wistuba II, Minna JD. "Essential role of aldehyde dehydrogenase 1A3 for the maintenance of non-small cell lung cancer stem cells is associated with the STAT3 pathway." Clin Cancer Res. 2014 Aug 1;20(15):4154-66. Epub 2014 Jun 6. http://dx.doi.org/10.1158/1078-0432.CCR-13-3292 ; PubMed PMID: 24907115; PubMed Central PMCID: PMC4438754 , Aug-2014
Articles in Peer-reviewed Journals	Su F, Mukherjee S, Yang Y, Mori E, Bhattacharya S, Kobayashi J, Yannone SM, Chen DJ, Asaithamby A. "Nonenzymatic role for WRN in preserving nascent DNA strands after replication stress." Cell Rep. 2014 Nov 20;9(4):1387-401. http://dx.doi.org/10.1016/j.celrep.2014.10.025 ; PubMed PMID: 25456133 , Nov-2014
Articles in Peer-reviewed Journals	El-Ashmawy M, Delgado O, Cardentey A, Wright WE, Shay JW. "CDDO-Me protects normal lung and breast epithelial cells but not cancer cells from radiation." PLoS One. 2014 Dec 23;9(12):e115600. eCollection 2014. http://dx.doi.org/10.1371/journal.pone.0115600 ; PubMed PMID: 25536195; PubMed Central PMCID: PMC4275221 , Dec-2014
Articles in Peer-reviewed Journals	Sridharan DM, Asaithamby A, Bailey SM, Costes SV, Doetsch PW, Dynan WS, Kronenberg A, Rithidech KN, Saha J, Snijders AM, Werner E, Wiese C, Cucinotta FA, Pluth JM. "Understanding cancer development processes after HZE-particle exposure: roles of ROS, DNA damage repair and inflammation." Radiat Res. 2015 Jan;183(1):1-26. http://dx.doi.org/10.1667/RR13804.1 ; PubMed PMID: 25564719 , Jan-2015
Articles in Peer-reviewed Journals	Zhang L, Kim SB, Jia G, Buhemeida A, Dallol A, Wright WE, Fornace AJ, Al-Qahtani M, Shay JW. "Exome sequencing of normal and isogenic transformed human colonic epithelial cells (HCECs) reveals novel genes potentially involved in the early stages of colorectal tumorigenesis." BMC Genomics. 2015;16(Suppl 1):S8. http://dx.doi.org/10.1186/1471-2164-16-S1-S8 , Jan-2015
Articles in Peer-reviewed Journals	Zhu J, Su F, Mukherjee S, Mori E, Hu B, Asaithamby A. "FANCD2 influences replication fork processes and genome stability in response to clustered DSBs." Cell Cycle. 2015;14(12):1809-22. Published online: 17 Jun 2015. http://dx.doi.org/10.1080/15384101.2015.1036210 ; PubMed PMID: 26083937 , Jun-2015
Articles in Peer-reviewed Journals	Elkin SR, Bendris N, Reis CR, Zhou Y, Xie Y, Huffman KE, Minna JD, Schmid SL. "A systematic analysis reveals heterogeneous changes in the endocytic activities of cancer cells." Cancer Res. 2015 Nov 1;75(21):4640-50. Epub 2015 Sep 10. https://doi.org/10.1158/0008-5472.CAN-15-0939 ; PubMed PMID: 26359453; PubMed Central PMCID: PMC4802864 , Nov-2015